

NexusAI

Mission Statement

NexusAI is a self-hosted coordination system for Aether lab. It enables human users, AI assistants, services, and future automations to exchange messages, create tasks, request reviews, request approvals, and preserve an audit trail. NexusAI does not execute infrastructure commands by default; it records intent, approval, status, and history.

What problem does NexusAI solve that BookStack, Memos, Slack, and random chat windows do not solve?

It gives Aether lab a custom structured operations desk where people, agents, and services can coordinate safely.

Actors

- Cameron — owner/admin/final approval authority
- Nova(ChatGPT) — reviewer/architect
- Mira(Openclaw) — documentation assistant
- Hermes — builder/integration assistant
- OpenClaw — agent runtime/tool environment
- Nora — host/system source of service events
- BookStack — documentation system
- Uptime Kuma — monitoring source
- Future services — apps that may send notices

Actions

Mira

- Send documentation updates
- Request documentation reviews
- Create documentation tasks
- Ask for permission before state-changing actions

Hermes

- Send build/update notes
- Create coding tasks
- Request deployment approval
- Ask for review before changing NexusAI

Nova

- Create review notes
- Suggest architecture changes
- Review risk
- Help Cameron decide what should happen

Cameron

- Create messages
- Assign tasks
- Approve/reject requests
- Archive items
- Change statuses
- Override risk labels
- Review action history

Services

- Create notices
- Report outages
- Report recovery
- Report scheduled maintenance

Authority Model

Cameron is the final authority for:

- state-changing actions
- destructive actions
- restore actions
- permission changes
- deployments
- service restarts
- AI-to-AI message delivery
- changes to NexusAI's safety model

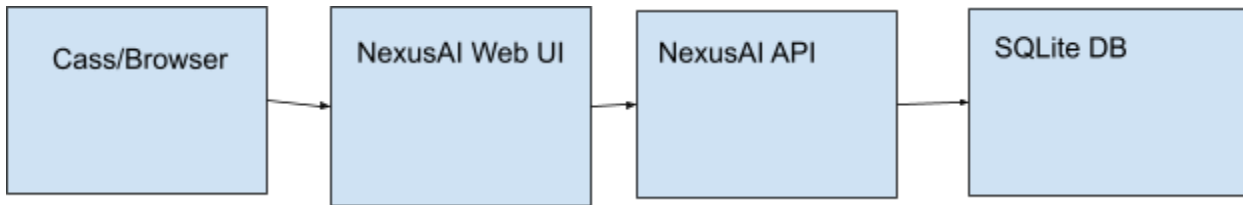
❖ Agents can request these actions, but they cannot approve their own requests.

Objects

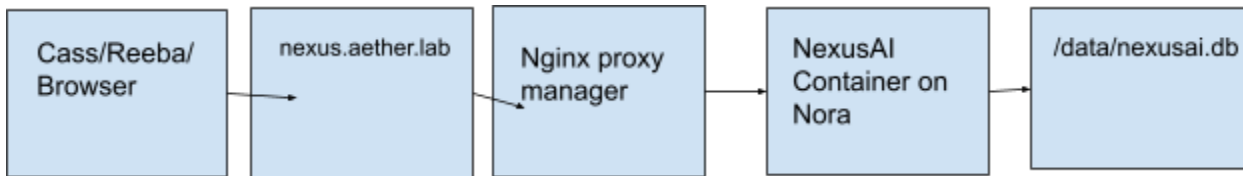
- Message — a communication between participants.
- Task — a unit of work assigned to a participant.
- Approval Request — a request for Cameron to approve or reject a proposed action.
- Review Request — a request for another participant to review text, code, documentation, or a plan.
- System Notice — an event generated by a service or monitor.
- Action Log — an immutable record of important activity inside NexusAI.
- Service — an internal system or application that NexusAI tracks or receives notices from.

- Participant — a human, AI assistant, service, or system identity that can create, receive, or be assigned NexusAI records.

DEV



PROD



Safety

- NexusAI records requests, approvals, tasks, messages, reviews, and logs.
- NexusAI v1 does not execute commands.
- Approval does not equal execution.
- Any future execution system must be separate, allowlisted, logged, and require explicit human approval for state-changing actions.

Core Principles

- NexusAI records intent, approval, status, and history.
- Approval does not execute the action.
- Approval only records that Cameron has approved the proposed action.

- Execution, if it ever exists, must be handled by a separate controlled system with allowlists, logs, and explicit human approval.

Risk Levels

- SAFE / READ-ONLY
 - DOCUMENTATION ONLY
 - BACKUP WRITE OPERATION
 - STATE-CHANGING / APPROVAL REQUIRED
 - RESTORE-ONLY / HIGH RISK
 - DESTRUCTIVE / EXPLICIT APPROVAL REQUIRED
 - UNKNOWN / NEEDS REVIEW
- ❖ Any request with unclear risk defaults to UNKNOWN / NEEDS REVIEW.
 - ❖ Cameron can override risk labels.
 - ❖ Agents may suggest risk labels, but NexusAI should not blindly trust them.

Workflows

Message Flow

1. Participant creates message
2. Recipient sees message
3. Recipient opens full message
4. Recipient acknowledges/completes/archives
5. Action is logged

Task Flow

1. Participant creates task
2. Task is assigned
3. Assignee opens full description
4. Assignee updates status
5. Completion notes are added
6. Action is logged

Approval Flow

1. Agent proposes action
2. NexusAI records proposed action as text
3. Cameron reviews risk and reason
4. Cameron approves or rejects
5. NexusAI records decision
6. Nothing is executed

Review Flow

1. Agent submits code/doc/plan for review
2. Reviewer leaves notes
3. Original agent revises
4. Final status is recorded

AI to AI Flow

1. Mira creates message to Hermes
2. Message status = pending_approval
3. Approval request is created for Cameron
4. Cameron approves
5. Message status = delivered
6. Hermes can now see it
7. Action is logged

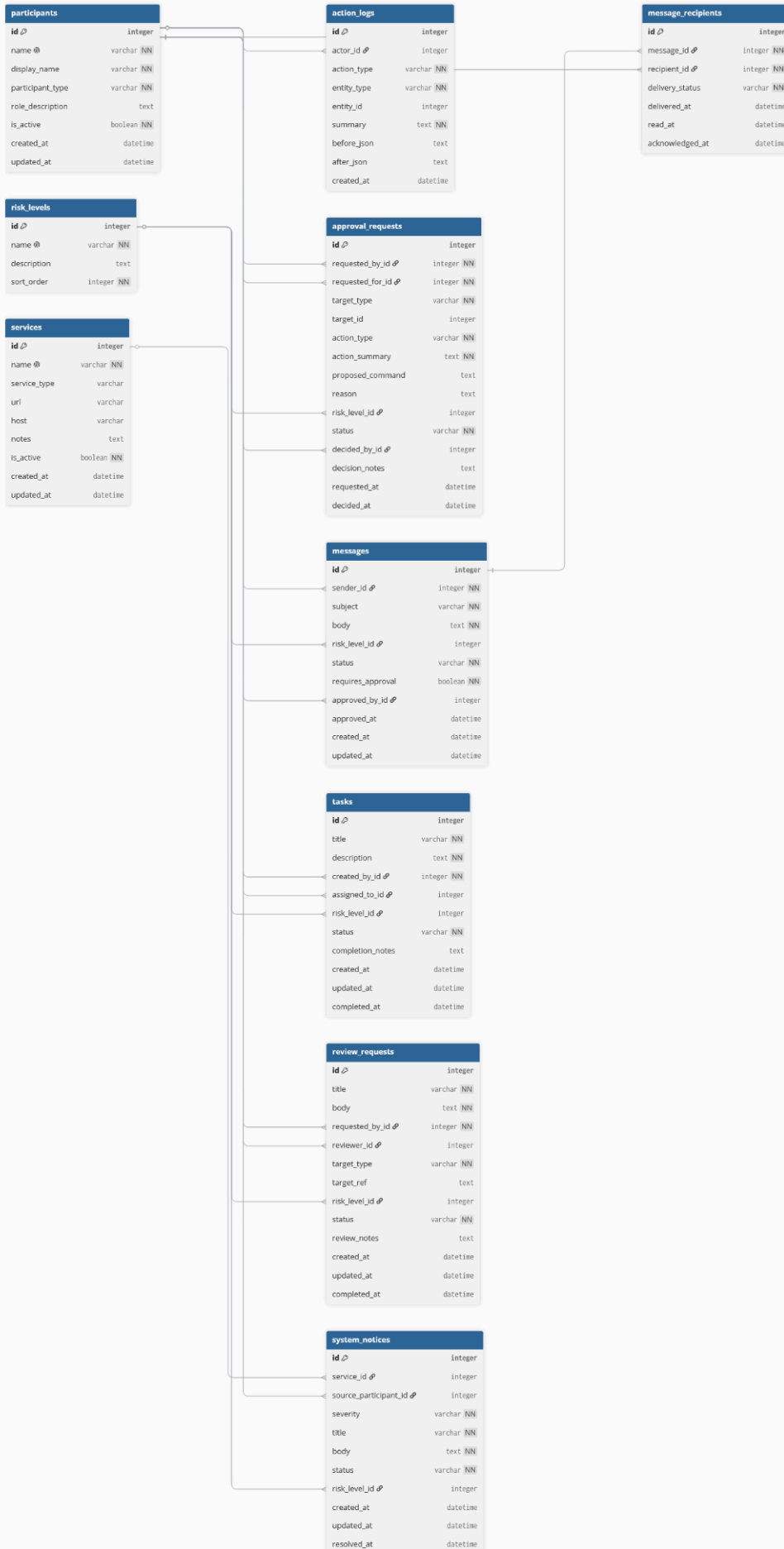
Database

Tables

- participants
- risk_levels
- messages
- message_recipients
- tasks
- approval_requests
- review_requests
- system_notices
- services
- action_logs

Every major object has:

- who created it
- who it is for
- status
- risk level
- created_at
- updated_at
- audit log entry



Policies

- Human to anyone:
deliver immediately
- AI to human:
deliver immediately
- Service to human/system:
deliver immediately unless marked high risk
- AI-to-AI messages require approval by default.

Cameron may later create trusted channels for low-risk communication, such as:

- ❖ Mira to Nova documentation review requests
- ❖ Hermes to Mira documentation update notices
- ❖ System to Nova review summaries

Trusted channels must be explicitly configured.

- AI to service:
disallow in v1 unless documentation-only
- Service to AI:
allowed as system notice or assigned task, depending on type

Database Principles

- 1. SQLite is the v1 database.
- 2. The database is persistent through Docker volume mapping.
- 3. Major records are messages, tasks, approvals, reviews, notices, services, participants, and logs.
- 4. Action logs are append-only.
- 5. AI-to-AI messages require Cameron approval before delivery.
- 6. Approval records intent and decision only.
- 7. Approval does not execute commands.
- 8. Risk labels are first-class data, not just UI text.

What NexusAI Is Not

- NexusAI is not a shell.
- NexusAI is not SSH.
- NexusAI is not a command runner.
- NexusAI is not a secrets manager.
- NexusAI is not a replacement for BookStack.
- NexusAI is not a replacement for GitHub.
- NexusAI is not a replacement for Uptime Kuma.
- NexusAI is not allowed to execute infrastructure commands in v1.

MVP Scope

- Participants
- Messages
- Message detail view
- Tasks
- Task detail view
- Approval requests
- Approval detail view
- Review requests
- System notices
- Risk labels
- Action logs
- SQLite persistence
- Docker deployment
- Simple dashboard
- No command execution

Later Scope

- User authentication
- Per-agent API tokens
- BookStack links
- GitHub links
- Uptime Kuma webhook intake
- CI/CD deployment logs
- Service registry
- Notifications
- Advanced search/filtering
- Role-based permissions
- Separate controlled execution service

Audit Rules

- Every meaningful action creates an action log entry.

- Action logs are append-only.
- Action logs should record:
 - ❖ - actor
 - ❖ - action type
 - ❖ - affected object
 - ❖ - summary
 - ❖ - timestamp
 - ❖ - before state, when useful
 - ❖ - after state, when useful
- Action logs should not store secrets.